KAIZEN MATHS

Lesson starter

Confidence Intervals

t-Intervals for a Mean

Name _____

SA1

Calculate a t confidence interval for a mean.

A sample of size 10 has mean 46.2 and sample standard deviation 5.2. Find a 95% t confidence interval for the population mean.

SA2

Find the t margin of error.

A sample has n = 10, sample standard deviation s = 10.2, and df = 9. Find the 95% t margin of error.

Show your working

KAIZEN MATHS

Lesson starter

Confidence Intervals

t-Intervals for a Mean

Name _____

SB1

Calculate and interpret a t confidence interval in context.

A teacher records the time taken by 25 students to complete a task. The sample mean is 53.5 minutes and the sample standard deviation is 4.5 minutes. Find and interpret a 90% confidence interval for the mean time.

SB2

Calculate a t confidence interval for a mean.

A sample of size 10 has mean 79.2 and sample standard deviation 6.4. Find a 99% t confidence interval for the population mean.

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KAIZEN MATHS

Exit ticket

Confidence Intervals

t-Intervals for a Mean

Name _____

EA1

Find the t margin of error.

A sample has n = 25, sample standard deviation s = 10.2, and df = 24. Find the 99% t margin of error.

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KAIZEN MATHS

Exit ticket

Confidence Intervals

t-Intervals for a Mean

Name _____

EB1

Calculate and interpret a t confidence interval in context.

A teacher records the time taken by 15 students to complete a task. The sample mean is 46.5 minutes and the sample standard deviation is 6.4 minutes. Find and interpret a 95% confidence interval for the mean time.

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Exit ticket

Confidence Intervals

t-Intervals for a Mean

Name _____

EC1

Calculate a t confidence interval for a mean.

A sample of size 25 has mean 38.8 and sample standard deviation 5.2. Find a 99% t confidence interval for the population mean.

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Confidence Intervals

t-Intervals for a Mean

Name _____

EA1

Find the t margin of error.

A sample has n = 25, sample standard deviation s = 10.2, and df = 24. Find the 99% t margin of error.

Show your working

Teacher answers & working

Confidence Intervals · t-Intervals for a Mean

STA-A06D-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SA1

Calculate a t confidence interval for a mean.

A sample of size 10 has mean 46.2 and sample standard deviation 5.2. Find a 95% t confidence interval for the population mean.

Answer:

95% t CI: (42.480, 49.920).

Working:

Because σ is unknown, use a t interval with df = n − 1 = 10 − 1 = 9.

For 95% confidence and df = 9, t = 2.262.

Standard error = $\frac{s}{\sqrt{n}} = \frac{5.2}{\sqrt{10}} = 1.6444$.

Margin of error = 2.262 × 1.6444 = 3.7196.

Interval = 46.2 ± 3.7196 = (42.480, 49.920).

SA2

Find the t margin of error.

A sample has n = 10, sample standard deviation s = 10.2, and df = 9. Find the 95% t margin of error.

Answer:

Margin of error = 7.296.

Working:

For 95% confidence and df = 9, t = 2.262.

Standard error = $\frac{s}{\sqrt{n}} = \frac{10.2}{\sqrt{10}} = 3.2255$.

Margin of error = t × standard error.

Margin of error = 2.262 × 3.2255 = 7.296.

Teacher answers & working

Confidence Intervals · t-Intervals for a Mean

STA-A06D-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

SB1

Calculate and interpret a t confidence interval in context.

A teacher records the time taken by 25 students to complete a task. The sample mean is 53.5 minutes and the sample standard deviation is 4.5 minutes. Find and interpret a 90% confidence interval for the mean time.

Answer:

90% CI: (51.960, 55.040) minutes.

Working:

Use a t interval because the population standard deviation is not known.

$\mathrm{df} = 25 - 1 = 24$.

t = 1.711 for 90% confidence.

Standard error = $\frac{4.5}{\sqrt{25}}$ = 0.9000.

Margin of error = 1.711 × 0.9000 = 1.5399.

Interval = 53.5 ± 1.5399 = (51.960, 55.040).

We are 90% confident that the population mean time lies between 51.960 and 55.040 minutes.

SB2

Calculate a t confidence interval for a mean.

A sample of size 10 has mean 79.2 and sample standard deviation 6.4. Find a 99% t confidence interval for the population mean.

Answer:

99% t CI: (72.622, 85.778).

Working:

Because σ is unknown, use a t interval with df = n − 1 = 10 − 1 = 9.

For 99% confidence and df = 9, t = 3.250.

Standard error = $\frac{s}{\sqrt{n}} = \frac{6.4}{\sqrt{10}}$ = 2.0239.

Margin of error = 3.250 × 2.0239 = 6.5775.

Interval = 79.2 ± 6.5775 = (72.622, 85.778).

Teacher answers & working

Confidence Intervals · t-Intervals for a Mean

STA-A06D-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EA1

Find the t margin of error.

A sample has n = 25, sample standard deviation s = 10.2, and df = 24. Find the 99% t margin of error.

Answer:

Margin of error = 5.706.

Working:

For 99% confidence and df = 24, t = 2.797.

Standard error = $\frac{s}{\sqrt{n}} = \frac{10.2}{\sqrt{25}} = 2.0400$.

Margin of error = t × standard error.

Margin of error = 2.797 × 2.0400 = 5.706.

EB1

Calculate and interpret a t confidence interval in context.

A teacher records the time taken by 15 students to complete a task. The sample mean is 46.5 minutes and the sample standard deviation is 6.4 minutes. Find and interpret a 95% confidence interval for the mean time.

Answer:

95% CI: (42.955, 50.045) minutes.

Working:

Use a t interval because the population standard deviation is not known.

$\mathrm{df} = 15 - 1 = 14$.

t = 2.145 for 95% confidence.

Standard error = $\frac{6.4}{\sqrt{15}} = 1.6525$.

Margin of error = 2.145 × 1.6525 = 3.5446.

Interval = 46.5 ± 3.5446 = (42.955, 50.045).

We are 95% confident that the population mean time lies between 42.955 and 50.045 minutes.

Teacher answers & working

Confidence Intervals · t-Intervals for a Mean

STA-A06D-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EC1

Calculate a t confidence interval for a mean.

A sample of size 25 has mean 38.8 and sample standard deviation 5.2. Find a 99% t confidence interval for the population mean.

Answer:

99% t CI: (35.891, 41.709).

Working:

Because σ is unknown, use a t interval with df = n − 1 = 25 − 1 = 24.

For 99% confidence and df = 24, t = 2.797.

Standard error = $\frac{s}{\sqrt{n}} = \frac{5.2}{\sqrt{25}}$ = 1.0400.

Margin of error = 2.797 × 1.0400 = 2.9089.

Interval = 38.8 ± 2.9089 = (35.891, 41.709).